9

Speed control in electrical drives

Motor control means that e.g. speed or sometimes position has to be controlled with high accuracy. As in all control problems, disturbance signals are present, which is the reason that closed loop control is needed at all. The most common disturbance is load variations, but parameter variations are also common, and the closed loop control system is used to compensate for such variations. In e.g. an industrial robot, the inertia can change rapidly whereas it varies slowly in a reeling machine. The electrical parameters, e.g. resistances and inductances, do also vary as a function of e.g. temperature.

Closed loop control is important not only to control a certain quantity, but also to limit it. Thus an important function of a current controller in an electrical machine is to limit the maximum possible current in a fault situation.

The control loops in an electrical drive are usually always connected in cascade. This concept does only work if the bandwidth increases from the outer to the inner loops. The innermost loops are thus the fastest and the outermost the slowest. This means that the position control loop only can work well if the speed control loop quickly responds to any change in the reference value. The setting of suitable control parameters can thus be simplified if the inner loops are regarded as infinitely fast, or can be modeled with a simple model.

In this chapter speed control is discussed, without any demand for the type of torque source. The actual motor can thus be of any kind without affecting the design of the speed control loop. The mechanical system is briefly repeated in section 9.1. In section 9.2 the basics of cascade control are repeated. In chapter 9.3 the fundamental characteristics of the PI controller is discussed and in chapter 9.4 this is applied to speed control.

9.1 The mechanical system

The torque produced by the electrical machine, the electromechanical energy converter, is denoted T_{el} whereas the load torque that is a result of friction etc. is denoted T_l. A rotating mass with the inertia J has a speed ω that is determined by the differential equation:

$$\frac{d}{dt}(J \cdot \omega) = T_{el} - T_l \tag{9.1}$$

The inertia J is not necessarily constant and thus the derivative can be written as:

$$\frac{d}{dt}(J \cdot \omega) = J \cdot \frac{d\omega}{dt} + \omega \cdot \frac{dJ}{dt} = T_{el} - T_l \tag{9.2}$$

There are several examples of time varying inertia, e.g. industrial robots, reeling machines for paper and spin-dryers. In this context however we will consider J as constant. In the discussion so far we have assumed that the mechanical system is stiff. That is not true in many larger dives, i.e. the torsion of the shaft must be considered. Such analysis is also left out of this context.

9.2 Cascade control

If it is assumed that the torque production is infinitely fast, i.e. the real torque responds to changes in the reference value without time delay, then a block diagram of a speed and position control system can be described as in figure Figure 9.1.

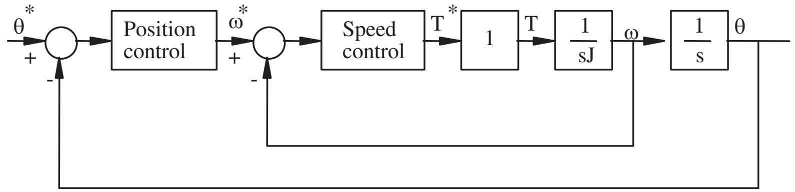

Figure 9.1: Block diagram for speed and position controllers.

The system contains two integrations. This gives a hint about two properties of the system:

- one must watch out for the stability margins;
- integrators may help to eliminate remaining errors.

The feedback in the speed loop has a directly stabilizing effect. To illustrate this, assume that both controllers are of proportional type with gain K_p for the position loop and K_ω for the speed loop. The closed system transfer function is then:

$$\frac{\theta(s)}{\theta^*(s)} = \frac{K_p \cdot K_\omega}{J \cdot s^2 + K_\omega \cdot s + K_p \cdot K_\omega} \tag{9.3}$$

with the poles:

$$poles = -\frac{K_\omega}{2J} \pm \sqrt{\frac{K_\omega^2}{4J^2} - \frac{K_p \cdot K_\omega}{J}}$$

The poles are always in the left half plane if K_{ϖ} and K_p are both positive. With these two parameters the poles of the position control system can be placed arbitrarily. Cascade control is thus nothing but a state feedback from the two states position and speed.

If we try to control the position with only one controller, i.e. the speed controller feedback is cut off and K_{ϖ}=1, the corresponding closed system transfer function will be:

$$\frac{\theta(s)}{\theta^*(s)} = \frac{K_p}{J \cdot s^2 + K_p} \tag{9.4}$$

The poles of this system are both on the imaginary axis, i.e. the system cannot be stabilized with only a proportional controller, it has to be derivative. However, derivating a position signal is equivalent of measuring the speed.

9.3 The PI controller

The PI controller can be used in most control loops in an electrical drive. There are reasons to take a closer look at this controller. It is available both in analogue and digital forms. At very high sampling frequency the digital controller resembles the analogue, but the implementation of anti windup functions are different.

Analogue PI controllers

The most commonly used parameterization of a PI controller uses the control error *e(t)*,

$$e(t) = y^*(t) - y(t) \tag{9.5}$$

where $y^*(t)$ is the reference value and $y(t)$ is the measured value. The controller output is proportional to the error and the integral of the error,

$$u(t) = K_p \cdot \left(e(t) + \frac{1}{\tau_i} \int e(t)dt \right) \tag{9.6}$$

It can also be written as,

$$u(t)=K_1 \cdot e(t)+K_2 \cdot \int e(t)dt \tag{9.7}$$

which theoretically is equal to equation (9.6), but from practical reasons the difference is big. In the equation (9.6) controller the gain can be changed without changing the time constant. This is important in commissioning.

The transfer functions with the controllers in equations (9.6) and (9.7) differ in another important sense as well. For (9.6) the transfer function can be written

$$u(s)=K_p \frac{1+s\tau_i}{s\tau_i} e(s) \tag{9.8}$$

This means that at a change in K_p the breakpoint in the Bode diagram will be unchanged $1/\tau_i$. The transfer function with controller according to (9.7) is

$$u(s) = \left(K_1 + \frac{K_2}{s\tau_i} \right) e(s) \tag{9.9}$$

As can be seen from (9.9) both gain and breakpoint will be affected by both K_1 and K_2.

Analog anti windup

If the control error *e(t)* remains for a longer time it is likely that the integral part becomes too large and saturates the control signal. In that case the integral part of the output signal must be limited. In a digital PI controller this can be done fairly easy by giving the integral part the value of the output (control) signal subtracted with the proportional part. In an analogue PI controller it is done with the use of two zener diodes in the feedback according to Figure 9.2

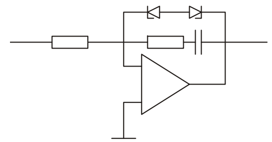

Figure 9.2: Anti windup connection of analogue PI controller.

Digital PI controllers

Digital PI-controllers can just like analog run into saturation of the output signal, i.e. limitation of the reference of the torque set point, and protection against "windup" must be introduced.

The digital PI controller is here assumed to operate at equidistant sampling instants denoted [k,k+1,k+2, …]. The output signal is built up of a proportional part and an integrating part according to:

$$u(k) = K_p \cdot \left(y^*(k) - y(k) + \frac{T_s}{T_i} \sum_{n=0}^{n=k} \left(y^*(n) - y(n) \right) \right) \tag{9.10}$$

where

$$K_p \cdot \frac{T_s}{T_i} \sum_{n=0}^{n=k} \left(y^*(n) - y(n) \right) = u_{\text{int}}(k) \tag{9.11}$$

and T_s is the sampling time of the controller. Thus the digital PI-controller can be expressed as:

$$u(k) = K_p \cdot \left(y^*(k) - y(k)\right) + u_{\text{int}}(k) \tag{9.12}$$

Note that the use of K_p in the calculation of the integral part of the controller has the same function as using equation (9.6) instead of (9.7) for the analog PI-controller, since a change of gain K_p will preserve the integral time constant.

$$u = K_p \frac{1 + \frac{T_s}{T_i} - z^{-1}}{\left(1 - z^{-1}\right)} \cdot e \tag{9.13}$$

Digital anti-windup

If there is a limitation in the control signal *u(k)* that may introduce windup of the integral part of the controller, an "anti windup" function can easily be introduced with equation (9.14).

$$\begin{aligned}
&u_{\text{int}}(k) = K_p \cdot \frac{T_s}{T_i} \sum_{n=0}^{n=k} \left(y^*(n) - y(n)\right) \\
&u(k) = K_p \cdot \left(y^*(k) - y(k)\right) + u_{\text{int}}(k) \\
&\text{if } u(k) > u_{\max} \text{ then} \\
&\quad u(k) = u_{\max} \\
&\quad u_{\text{int}}(k) = u_{\max} - K_p \cdot \left(y^*(k) - y(k)\right) \\
&\text{else if } u(k) < u_{\min} \text{ then} \\
&\quad u(k) = u_{\min} \\
&\quad u_{\text{int}}(k) = u_{\min} - K_p \cdot \left(y^*(k) - y(k)\right) \\
&\text{end if}
\end{aligned} \tag{9.14}$$

9.4 Adjustment of the speed controller

Assuming that the torque source is infinitely fast, the speed controller can be sketched as a block diagram in Figure 9.3. With a PI controller, the system will contain two integrators. It is possible to place the poles of that controller arbitrarily. Assume that the speed controller has the gain K and integration time constant T_i.

The closed system transfer function will then be

$$\frac{\omega(s)}{\omega^*(s)} = \frac{K\left(s \cdot T_i + 1\right)}{JT_i s^2 + KT_i s + K} \tag{9.15}$$

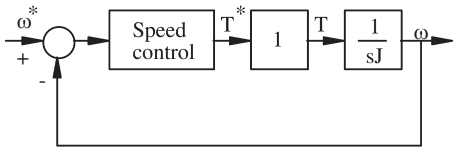

Figure 9.3: The speed control system.

The poles to the system will be

$$poles = -\frac{K}{2J} \pm \sqrt{\frac{K^2}{4J^2} - \frac{K}{JT_i}} \tag{9.16}$$

It can be seen from equation ((9.16) that the poles of the system can be placed arbitrarily with K and T_i. There are of course practical limitations to how fast the control system can be made. If K is to large, the assumption that the torque loop is infinitely fast are no longer valid. Furthermore the measured speed signal has to be filtered. If the toque loop dynamics are includes in the calculations a method called symmetric optimum can be used.

Symmetric optimum

A criterion often used in drives is "symmetric optimum", which is a standard criterion for circuits containing a double integrator. The idea is to select the cross over frequency ω_0 (when $|G(j\omega)| = 1$) at the frequency where the phase margin is maximal. This will give the best damping for the closed system

The actual torque response for a power electronically fed electrical machine is rather complicated to model. A first order approximation that is very useful is to assume that the actual torque responds to a change in the reference as a first order low pass filter with a time constant equal to the sampling period in the torque control loop.

$$\frac{T(s)}{T^*(s)} = \frac{1}{1 + s \cdot T_{tc}} \tag{9.17}$$

where T_{tc} is the band width of the torque control system. The speed control system can now be described as

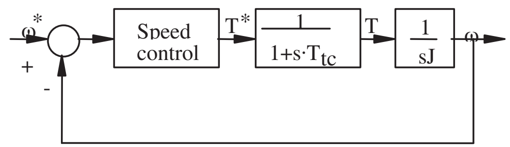

Figure 9.4: Block diagram for the speed controller, with a first order approximation of the torque loop dynamics included.

The open loop transfer function for the system in Figure 9.4 is,

$$G(j\omega) = K_\omega \frac{1+s\cdot T_i}{s\cdot T_i}\cdot\frac{1}{1+s\cdot T_{tc}}\cdot\frac{1}{s\cdot J} \tag{9.18}$$

We assume that $T_{tc} < T_i$. Then the Bode diagram for *G(j•)* has the principal behavior in Figure 9.5.

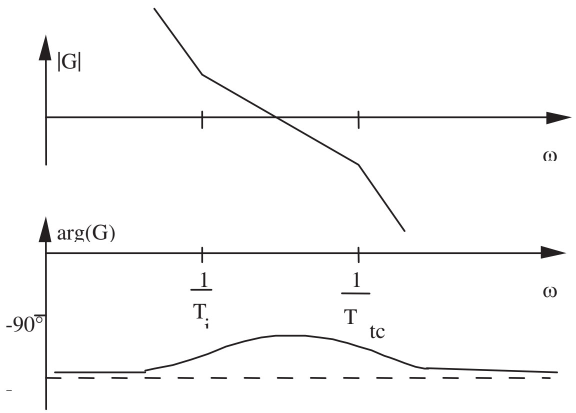

Figure 9.5: The principal behavior of *G (jω)* with PI controller, torque dynamics and mechanics included.

The maximum phase appears between the two breaking frequencies $1/T_i$ and $1/T_{tc}$

$$\omega_0 = \frac{1}{\sqrt{T_i\cdot T_{tc}}} \tag{9.19}$$

To determine the parameters of the PI controller, two equations are needed. The condition $|G(j\omega)| = 1$ at the frequency ω_0 gives the first equation. Assuming that $T_{tc} < T_i$:

$$T_i = a^2\cdot T_s\text{, where } a > 1$$

$$\omega_0 = \frac{1}{\sqrt{T_i\cdot T_{tc}}} = \frac{1}{a\cdot T_{tc}} = \frac{a}{T_i} \tag{9.20}$$

At the cross over frequency it is thus valid that

$$|G(j\omega)| = \left| K_\omega \frac{1+j\cdot a}{j\cdot a}\cdot\frac{1}{1+\frac{j}{a}}\cdot\frac{1}{\frac{j\cdot a}{T_i}\cdot J}\right| = K_\omega\cdot\frac{T_i}{a\cdot J} = 1$$

or

$$K_\omega = \frac{a\cdot J}{T_i} \tag{9.21}$$

The closed systems poles can be calculated explicitly. The transfer function for the closed speed loop is:

$$\frac{\omega(s)}{\omega^*(s)} = \frac{K_\omega \frac{1+s\cdot T_i}{s\cdot T_i}\cdot\frac{1}{1+s\cdot T_{tc}}\cdot\frac{1}{s\cdot J}}{1+K_\omega \frac{1+s\cdot T_i}{s\cdot T_i}\cdot\frac{1}{1+s\cdot T_{tc}}\cdot\frac{1}{s\cdot J}} = \frac{K_\omega(1+s\cdot T_i)}{s\cdot T_i\cdot(1+s\cdot T_{tc})\cdot s\cdot J + K_\omega(1+s\cdot T_i)} = \tag{9.22}$$
$$= \frac{K_\omega(1+s\cdot T_i)}{s^3\cdot J\cdot T_i\cdot T_{tc} + s^2\cdot J\cdot T_i + s\cdot K_\omega\cdot T_i + K_\omega}$$

With K_ω (9.21) and the relations (9.19), (9.20) inserted the characteristic equation can be written as

$$s^3\cdot\frac{T_i^2}{a^2} + s^2\cdot T_i + s\cdot a + \frac{a}{T_i} = 0 \tag{9.23}$$

One root is

$$s = -\omega_0 = -\frac{a}{T_i} \tag{9.24}$$

Polynomial division of the characteristic polynomial ((9.23) gives

$$s^2\cdot\frac{T_i^2}{a^2} + s\cdot T_i\cdot\frac{a-1}{a} + 1 = 0 \tag{9.25}$$

and then the other roots can be calculated analytically. We assume that they are complex conjugates,

$$s_{2,3} = -\omega_0\left(\xi \pm \sqrt{\xi^2 - 1}\right) \tag{9.26}$$

where

$$\zeta = \frac{a-1}{2} \tag{9.27}$$

and ω_0 is given by equation (9.19). We can see that ζ is the relative damping for poles to the system. The parameter T_i can now be selected to place the poles in a suitable way. If the relative damping is selected to $1/\sqrt{2}$, then a= 2.41. This gives directly a value on T_i as a function of T_{tc}. If we on the other hand assume that no complex poles are to exist in the speed control system that means that $\zeta = 1$, corresponding to $a = 3$. All three poles will thus be equal to $-\omega_0$.

Compensation for sensor noise

In most applications with speed control a speed sensor is needed, and the output signal of this speed sensor is usually subjected to noise that needs to be filtered out to some extent. Thus, the feed back loop involves a filter that affects the dynamics of the closed loop system. As shown above the characteristic equation of the closed loop is of the 3'rd order even without a filter, and with the filter included it would be of the 4'th order, thus rather difficult to find a useful algebraic solution.

A feasible engineering solution to this problem can be found by comparing the filter time constants of the speed sensor filter with the response time constant of the torque loop. The torque loop is usually very fast, expressed as a first order time constant the real torque follows the torque reference within 50…500 microseconds. The speed sensor filter time constant on the other hand is usually at least one and maybe two orders of magnitude longer, i.e. up to 50 milliseconds. To realize the impact of this on the control system parameter selection, a block diagram of the control system needs to be drawn. First it must be noted that it is not the speed that is controlled, but the speed measurement signal after the filter. Note from Figure 9.6 that from a control point of view the torque dynamics and the filter dynamics are series connected.

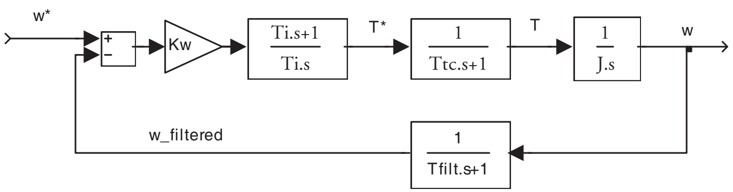

Figure 9.6: Speed control system with speed sensor filter dynamics.

Since the torque dynamics are at least one order of magnitude faster than the filter dynamics it can be omitted. With the torque dynamics omitted and the filter dynamics moved to structurally replace the torque dynamics it is evident that the system can be designed using symmetric optimum in the same way as in the previous section, but with the torque dynamics replaced by the filter dynamics.